

ADDIS ABABA UNIVERSITY

COLLEGE OF TECHNOLOGY AND BUILT ENVIRONMENT (COTBE)

SCHOOL OF ELECTRICAL AND COMPUTER ENGINEERING

(COMPUTER STREAM)

… Impact of Social Media on Academic Performance…

Group Members

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1 Abstract

Title: *The Impact of Social Media on Academic Performance Among University Students*

This study investigates the relationship between social media usage and academic performance, focusing on the students starting from High school. With the pervasive use of platforms like Instagram, TikTok, and Facebook, concerns about their effects on study habits and grades have grown. The primary objective is to determine whether excessive social media use correlates with lower GPAs, while also exploring platform-specific impacts.

A **quantitative correlational design** will be employed, surveying 200 students from highschool and universities. Data will be collected via a structured questionnaire measuring daily usage time, preferred platforms, and self-reported GPA. **Statistical analysis** (Pearson's r, ANOVA) will identify significant relationships, controlling for variables like major and study hours.

Expected outcomes include:

1. A negative correlation between >2 hours/day of social media use and GPA.
2. TikTok/Instagram users may show stronger performance declines than LinkedIn users.
3. Gender differences in usage patterns and academic impact.

The findings aim to guide educational interventions, such as time-management workshops, and contribute to the debate on digital literacy in academia. Limitations include reliance on self-reported data and regional sampling.

2 Introduction

Context of the Study

Social media platforms (e.g., Instagram, TikTok, Snapchat) have become integral to daily life, especially among students. Recent data shows 95% of high schoolers and 89% of university students use social media daily, averaging 2.5–4 hours per day (Pew Research, 2023). While these platforms offer connectivity, growing evidence suggests excessive use may disrupt academic focus, sleep patterns, and memory retention — critical factors for learning outcomes.

This study expands on prior research by examining both high school (ages 13–18) and university students (ages 18–25), addressing gaps in two key areas:

1. Developmental Differences: Does social media impact high schoolers' grades more severely due to evolving self-regulation skills?
2. Platform-Specific Effects: Do short-form video apps (TikTok) correlate with poorer performance than text-based platforms (Twitter)?

Relevance and Motivation

1. Educational Policy: Schools globally are debating social media bans during class hours (e.g., Florida's 2024 classroom ban). Empirical data is needed to guide policies.
2. Parental Concerns: 68% of parents worry about social media's academic impact (Common Sense Media, 2023). This study provides actionable insights.
3. Global Relevance: Cross-cultural data from Ethiopia will complement Western-centric studies.

3 Statement of the Problem

Despite global studies on social media's academic impact, critical gaps persist in Ethiopia's educational context:

- Limited Local Data: Most research focuses on Western/Asian students (e.g., Alghamdi 2022), neglecting African adolescents' unique socio-technological landscape.
- Infrastructure Disparities: Ethiopia's internet penetration (~25% as of 2023) and frequent outages may alter usage patterns compared to global averages.
- Cultural Nuances: High schoolers in Ethiopia often balance social media with familial obligations (e.g., household duties), potentially intensifying time-management challenges.

Research Gap:

No longitudinal studies in Ethiopia examine how platform-specific use (e.g., Telegram for study groups vs. TikTok for entertainment) correlates with standardized exam scores (e.g., Ethiopian Higher Education Entrance Certificate).

4. Literature Review

Global Perspectives

1. **Negative Correlation**
    - A meta-analysis of 32 studies (Hawi & Nassar, 2023) found that >2 hours/day of social media use reduced GPA by 0.5 points, with teens (13–19) more affected than adults.
    - *Gap*: Only 2% of participants were from Africa.
2. **Platform-Specific Effects**
    - TikTok/Instagram users scored 15% lower on memory tasks than LinkedIn users (University of Oxford, 2023).
    - *Gap*: Studies focused on high-income countries with stable internet.

African Context

3. **Ethiopian Studies**
    - A survey at Addis Ababa University (Tesfaye, 2022) linked Facebook use to disrupted sleep (67% of respondents), but did not measure academic performance.
    - *Gap*: No data on high school students or rural areas.

4. **Infrastructure Challenges**
    - Research in Kenya (Mugo, 2023) showed that intermittent internet access increased anxiety around academic use of social media.
    - *Relevance to Ethiopia*: Similar connectivity issues likely amplify this effect.

Theoretical Framework

- **Social Cognitive Theory** (Bandura, 1986): Explains how Ethiopian students may imitate peers' procrastination behaviors (e.g., joining "all-night" Telegram study groups that devolve into chatting).
- **Uses & Gratifications Theory**: Suggests Ethiopian teens may prioritize utilitarian SM uses (e.g., accessing textbooks via WhatsApp) over entertainment compared to Western peers.

5. Hypotheses/Research Questions

Quantitative Hypotheses

1. **Primary Hypothesis**:
    - There is a negative correlation between daily social media usage (>2 hours) and self-reported academic performance (GPA/exam scores) among Ethiopian high school and university students ($p < .05$).
2. **Secondary Hypotheses**:
    - Students using entertainment-focused platforms (TikTok, Instagram) will report lower GPAs than those using education-focused platforms (Telegram study groups, LinkedIn) (*α = .05*).
    - High school students (ages 13–18) will show a stronger negative correlation than university students (ages 18–25) due to less developed self-regulation.

Qualitative Guiding Questions

1. How do Ethiopian students perceive social media's role in their study habits? *(Thematic analysis of interviews)*
2. What coping strategies do high-achieving students employ to balance social media and academics? *(Case studies)*

6. Conceptual Framework

Relationship Between Social Media Usage and Academic Performance



Explanation of Variables:

1. **Independent Variable (IV):** Social Media Usage

- Measured by:
    - Daily hours spent on platforms (Facebook, Instagram, TikTok, X/Twitter, etc.)
    - Purpose (entertainment, networking, academic use)
    - Frequency of checking notifications

2. **Dependent Variable (DV):** Academic Performance

- Measured by:
    - GPA (Grade Point Average)
    - Exam scores
    - Assignment submission timeliness
    - Class participation

3. **Mediating/Moderating Variables:**

- Time Management: Excessive social media may reduce study time.
- Sleep Disruption: Late-night usage affects cognitive function.
- Attention Span: Frequent interruptions reduce concentration.

7. Objectives and Aim of the Study

The general aim of this study is to investigate the relationship between social media usage and academic performance among students ranging from high school to university level. The study seeks to explore the extent and nature of students' engagement with social media platforms and how this engagement influences their academic outcomes.

Specifically, the study aims to:

- Determine the average daily time students spend on social media.
- Assess the correlation between social media engagement and students' Grade Point Averages (GPAs).
- Identify the social media platforms that are most distracting to students.
- Evaluate how social media usage affects students' study habits, including procrastination and multitasking.
- Compare academic performance between high-frequency and low-frequency social media users.
- Explore gender-based differences in how social media impacts academic performance.
- Propose practical recommendations for promoting balanced and responsible social media use among students.

8. Research Methods, Materials, and Procedures

8.1 Study Area

This research was conducted in educational institutions located in Addis Ababa, the capital city of Ethiopia. The study focused on students aged between 14 and 25 years, encompassing both high school and undergraduate university students. The population represents diverse socioeconomic backgrounds, allowing for a culturally rich context in which to explore the research questions.

8.2 Study Design

A mixed-methods research design was employed, combining both quantitative and qualitative approaches to capture a holistic understanding of the phenomenon. The study adopted a correlational framework, aiming to examine associations between social media usage and academic performance without manipulating any variables. Quantitative data were collected through structured survey questionnaires incorporating Likert-scale items, while qualitative insights were gathered through focus group discussions to explore participants' experiences and perceptions in greater depth.

8.3 Study Subjects

The study targeted students enrolled in high schools and undergraduate university programs. The population was stratified into two groups for comparative analysis: Group 1 consisted of high school students aged 14 to 18 years, and Group 2 included undergraduate students aged 18 to 22 years.

8.4 Eligibility Criteria

To ensure the validity of the findings, specific inclusion and exclusion criteria were established. Participants were eligible for inclusion if they were currently enrolled in an academic institution and used social media for at least one hour per day. Students who did not use social media or who had diagnosed learning disabilities were excluded from the study to avoid potential confounding variables that could impact academic performance independently of social media use.

8.5 Sample Size

For the quantitative component of the study, a power analysis was conducted to determine the appropriate sample size. Assuming a 95% confidence level, a 5% margin of error, and a medium effect size (Cohen's d = 0.5), the minimum required sample size was calculated to be 384 students, using the G*Power statistical tool. For the qualitative portion, thematic saturation was used as the guiding principle, with 15 to 20 participants included until no new themes emerged from the data.

8.6 Sampling Methods

A stratified random sampling technique was utilized to ensure that the sample was representative across various demographic groups. Stratification was based on gender (ensuring an equal representation of male and female students) and type of institution (public versus private). The procedure began with obtaining a list of eligible schools and universities in Addis Ababa. From these, classes were randomly selected, and students within those classes were invited to participate voluntarily after providing informed consent.

8.7 Method of Data Collection

To ensure a comprehensive and Good understanding of the relationship between social media usage and academic performance, we adopt a mixed methods and approach utilizing both quantitative and qualitative data collection tools.

8.7.1 Survey Questionnaire (Quantitative)

A structured questionnaire was developed based on the conceptual framework. It included:

- **Closed-ended questions** using a 5-point Likert scale (e.g., strongly agree to strongly disagree).
- **Multiple-choice items** about time spent on platforms, purposes of use, and academic indicators.
- Administered both **online (Google Forms)** and **offline (paper forms)** to reach students in schools and universities in Addis Ababa.

We distributed forms at selected institutions and coordinated online sharing through school group chats, email, and Telegram.

8.7.2 Focus Group Discussions (Qualitative)

To get deeper insights into the effects of social media, we conducted 2 small focus group discussions:

- Each group had 5–8 volunteer students from different grade levels.
- Discussions were semi-structured and focused on themes like procrastination, sleep habits, and concentration.

8.8 Description of Data Variables

Variable	Type	Description / Measurement
Social Media Usage	Independent Variable	Daily hours, purpose (academic/social/entertainment), notification frequency
Academic Performance	Dependent Variable	Self-reported GPA, test scores, class participation, and assignment timeliness
Time Management	Mediating Variable	“I delay studying because of social media” (Likert scale)
Sleep Disruption	Mediating Variable	“I stay up late due to social media”
Attention Span	Mediating Variable	“I get distracted while studying because of notifications”
Gender, Institution Type	Stratification Criteria	For comparisons across demographics

8.9 Data Quality Assurance

To ensure the accuracy and reliability of the data , we used the following steps:

- **Pilot test**: We pre-tested the survey on 20 students to ensure clarity , Ambiguous or misleading questions were revised.
- **Clear instructions** were included to help participants answer correctly and completely.
- **Anonymous responses** were collected to encourage honesty and reduce social desirability bias.
- We manually **checked and cleaned the data** for incomplete or contradictory answers before analysis.

8.10 Operational Definitions

- **Social Media Usage**: The total amount of time a student spends per day on platforms like TikTok, Instagram, Facebook, and Telegram, as well as the reasons for usage like entertainment, communication, academic tasks.

- **Academic Performance**: Includes GPA (self-reported), exam scores, timely assignment submission, and active class participation.

- **High-Frequency User**: A student who spends more than 4 hours daily on social media.

- **Low-Frequency User**: A student who spends less than 2 hours daily.

- **Distraction**: When checking or using social media interrupts studying or classroom focus.

8.11 Plan of Data Analysis

Given our student-level resources, we use Microsoft Excel, Google Sheets, and simple statistical tools for data analysis.

8.11.1 Quantitative Data (Survey):

- **Descriptive Statistics**: Frequencies, percentages, and means to describe usage patterns and academic performance.

- **Comparative Analysis**: Using t-tests or cross-tabulation to compare GPA between high and low users.

- **Correlation Analysis**: To assess relationships between variables like hours spent and GPA or sleep disturbance and academic scores.

8.11.2 Qualitative Data (Focus Groups):

- Responses are coded manually to identify common patterns.

- We'll group responses by theme (e.g., procrastination, anxiety, focus issues) to enrich the numerical data with lived experiences.

9. Work Plan (Timeline)

The project will span 10 weeks, considering the academic semester constraints. Below is a table outlining the timeline:

Task	**Week 1**	**Week 2**	**Week 3**	**Week 4**	**Week 5**	**Week 6**	**Week 7**	**Week 8**	**Week 9**	**Week 10**
Literature Review	X	X								
Design Online Survey		X	X							
Distribute Survey			X	X	X					
Data Analysis					X	X	X			
Draft Report							X	X		
Final Review & Submission								X	X	X

Contingency Plan (If Timeline Is Not Met)

- **Literature Review Delay:** If not completed by Week 2, reduce scope to key articles and finalize by Week 3 while starting survey design in parallel.
- **Survey Distribution Delay:** If delayed beyond Week 5, extend distribution by 1 week, using university email lists and social media groups to accelerate responses.
- **Data Analysis Delay:** If not completed by Week 7, use simpler statistical methods (e.g., descriptive statistics) to ensure report drafting begins on time.
- **Report Drafting Delay:** If delayed, team members will divide sections and work concurrently, submitting a draft for review by Week 9.

10. Budget (Estimated Costs)

As a university student in Ethiopia, costs are minimized by using free or low-cost resources. Expenses are in Ethiopian Birr (ETB).

Item	**Estimated Cost (ETB)**
Internet data for online surveys (e.g., Ethio Telecom bundles)	400 ETB
Printing (final report copies for submission)	300 ETB
Stationery (notebooks, pens for planning)	200 ETB
Total	**900 ETB**

11. References

References are formatted in APA style:

·Brown, L., & Green, M. (2023). Social media addiction and its effects on GPA: A quantitative analysis. *Educational Research Review*, 40, 101–115. https://doi.org/10.1016/j.edurev.2023.101

· Desta, M., & Kebede, S. (2024, March). Balancing social media and online learning: Perspectives from engineering students. *Proceedings of the International Conference on Education Technology, Addis Ababa, Ethiopia*, 112–120.

·Johnson, K. L., & Amare, T. (2022). The impact of social media usage on academic performance: A study of university students in Ethiopia. *Journal of Educational Technology & Society*, 25(3), 45–58.

·Smith, J. A. (2021). *Digital distractions: Social media and student success*. Academic Press.

·Taylor, R., & Patel, N. (2020). Patterns of social media use among university students: A cross-sectional study. *Computers in Human Behavior*, 105, 106–119. https://doi.org/10.1016/j.chb.2020.106

12. Appendices/Annexes

Appendix A: Online Survey Questionnaire

Online Survey on Social Media Usage and Academic Performance
Distributed via Google Forms

1. What is your age (years)?
   - ➔ <15
   - ➔ 15–18
   - ➔ 18–25
   - ➔ 25–35
   - ➔ >35
2. What is your gender?
   A Male B Female
3. Which course(s) are you currently enrolled in?
   A PG C High school
   B UG D Other
4. Current academic year?
   A Grade 9 – 10 C Year 1-2
   B Grade 11- 12 D Year 3-4 E ≥Year 4
5. Do you use social media for academic purposes?
   A Yes B No C Not sure
6. Service(s) most frequently used by you?
   A Facebook E Telegram
   B Tik Tok F WhatsApp
   C Youtube G LinkedIn
   D Google+ H Other
7. How long have you been using such services?
   A <1 year C 1–3 years
   B 3-5 years D >5 years
8. Time spent on social media daily?
   A <1 h C 1–3 h
   B >3 h
9. Approximate monthly expenditure spent on social media usage (ETB)?

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10. Do you feel addicted to social media?
    A Yes B No C Not sure
13. Has social networking impacted your academic performance?
    A Yes B No C Not sure
14. What was the nature of the impact?
    A Improved C No effect

B Worsened D Not sure

15. What is your current Average/CGPA?

A < Avg 50 D Avg 50 Avg 80

B > Avg 80 E Below 2.0

C 2.0–3.0 F Above 3.0

Appendix B: Digital Consent Statement

Digital Consent Statement for Participation in Research Study

This study, titled "Impact of Social Media on Academic Performance," is being conducted anonymously via Google Forms. No personally identifiable information will be collected, and your responses will be used solely for research purposes. Participation is voluntary, and you may withdraw at any time by exiting the survey. By proceeding with the survey, you confirm that you understand the purpose of the study and agree to participate.

I have read the above statement and agree to participate

Appendix C: Raw Data Template

To be exported from Google Forms in CSV format

Response ID	**Age**	**Gender**	**Course**	**Academic Year**	**Social Media for Academic (Y/N)**	**Most Used Service**	**Daily Time on Social Media**	**Monthly Expenditure (ETB)**	**Feel Addicted (Y/N)**	**Impact on Performance (Y/N)**	**Nature of Impact**	**GPA**
001												
002												